

Course Schedule — 15 Weeks, 30 Lectures

Discrete-Event Simulation: Modeling and Simulation in the AI Era

See syllabus for policies

Lecture Plan

Wk	Lec	Title	Ch.	Mod.	Assignment
1	L01	Why Simulate? History and Applications	1	M01	
1	L02	A Taxonomy of Simulation Models	2	M01	HW-01 out
2	L03	System Boundaries, Entities, Resources	3	M02	
2	L04	Events, State Variables, Performance Measures	3	M02	
3	L05	The Simulation Clock and Event Calendar	4	M03	HW-01 due
3	L06	The Event Loop and Process-Oriented Worldview	4	M03	HW-02 out
4	L07	Manual Tracing: Single-Server Queue	5	M03	
4	L08	Little's Law and Performance Estimation	5	M03	
5	L09	Python and SimPy Essentials	6	M04	HW-02 due
5	L10	The <code>simdes</code> Package and BaseModel Interface	6	M04	HW-03 out
6	L11	The M/M/1 Queue in SimPy	7	M05	
6	L12	The M/M/c Queue: Erlang-C	7	M05	
7	L13	M/G/1 and Variance Effects	7	M05	HW-03 due
7	L14	Finite-Capacity and Priority Queues	7	M05	HW-04 out
8	L15	Clinic Case Study: Phases 1–2	7	M05	
8	L16	Verifying Simulation Against Theory	7	M05	
9	L17	Non-Stationary Arrivals and Routing	8	M06	HW-04 due
9	L18	Batching, Stores, and Containers	8	M06	
10	L19	Interrupts and Preemption	8	M06	

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Wk	Lec	Title	Ch.	Mod.	Assignment
10	L20	Multi-Stage and Network Models	8	M06	
11	L21	The (s, S) Inventory Control Model	8	M06	
11	L22	(s, S) Scenario Analysis	8	M06	HW-05 out
12	L23	Input Modeling: EDA and Distribution Fitting	9	M07	
12	L24	Goodness-of-Fit Tests	9	M07	Midterm out
13	L25	Input Modeling: Scarce Data and Expert Knowledge	9	M07	HW-05 due
13	L26	Warm-Up Detection: Welch Method	10	M08	HW-06 out
14	L27	Independent Replications and Confidence Intervals	10	M08	
14	L28	Common Random Numbers and Scenario Comparison	10	M08	
15	L29	Experimental Design: 2^k Factorials	11	M09	HW-06 due
15	L30	Sensitivity Analysis and Ranking/Selection	11–12	M09–M10	Final out

Midterm project due: End of Week 14.**Final project due:** Finals week (see course calendar).

Advanced Topics (Weeks 13–15 Extension)

The following topics are integrated into the schedule as time permits, or as separate seminar sessions for graduate credit / advanced tier students.

Module	Chapter	Topic
M10	Ch. 12	Verification and Validation
M11	Ch. 13	Simulation Optimization (Weeks 13–14 extension)
M12	Ch. 14	Reinforcement Learning for Simulation Control (Week 15 extension / final project option)

Assignment Calendar

Assignment	Out	Topic	Due
HW-01	L02	Conceptual Modeling	L05
HW-02	L06	Manual Simulation Trace	L09
HW-03	L10	Queue Models	L13
HW-04	L14	Advanced DES	L17
HW-05	L22	Input Modeling	L25
HW-06	L26	Output Analysis	L29
HW-07	L28	Experimental Design	End of Week 14
Midterm Project	L24	Full simulation study	End of Week 14
Final Project	L30	Advanced simulation study	Finals week